

DAY-14

(13-06-26)

⇒ Arithmetic Operations:-

① Decimal Addition:-

(a) $[23]_{10}$

$(+)$ $[72]_{10}$
 $[95]_{10}$

(b)

74
 $+ 88$
 $(162)_{10}$
 $74 + 8 = 12$
 $12 - 10 = 2$

(★) When you cannot write a number, Subtract it from its base & Write

② Decimal Subtraction:-

(a) 52

$- 21$
 31

(b) 74

36
 $- 38$

(★) Add it to base & Write.

⇒ standard calculations in binary number systems is important. For other bases, we can do calculation by converting it to decimal.

Ex:- ① $(227)_8 + (562)_8 = (?)_8$

M-I:- $(227)_8 = 7 \times 8^0 + 2 \times 8^1 + 2 \times 8^2 = 7 + 16 + 128 = (151)_{10}$
 $(562)_8 = 2 \times 8^0 + 6 \times 8^1 + 5 \times 8^2 = 2 + 48 + 320 = (370)_{10}$
 $(521)_{10}$

⇒ $(521)_{10} = (?)_8$

$8 \overline{) 521}$
 $8 \overline{) 65}$
 $8 \overline{) 81}$
 1
 0

$= (1011)_8$

M-II:-

117
 $+ 562$
 $(1011)_8$

$7 + 2 = 9 - 8 = 1$

$6 + 2 + 1 = 9 - 8 = 1$

$5 + 2 + 1 = 8 - 8 = 0$

② $(1.2)_4 + (2.3)_4 = (?)_2$

~~$(1.2)_2 + (2.3)_2 = (?)_2$~~

~~$(0.11)_2 + (1.11)_2$~~

M-I:-

$(1.2)_4 = 1 \times 4^0 + 2 \times 4^{-1} = 1 + 0.5 = (1.5)_{10}$

$(2.3)_4 = 2 \times 4^0 + 3 \times 4^{-1} = 2 + 0.75 = (2.75)_{10}$

⇒ $1.5 + 2.75 = (4.25)_{10} = (?)_2$

4.25
 $(4.25)_{10}$

$2 \overline{) 4.25}$
 $2 \overline{) 4}$
 1
 0

$0.25 \times 2 = 0.50$

$0.50 \times 2 = 1.00$

$(100.01)_2$

M-II

$$\begin{array}{r} 1 \\ 1 \cdot 2 \\ + 2 \cdot 3 \\ \hline (10 \cdot 1)_4 \end{array}$$

$$3+2 = 5 - 4 = 1$$

$$2+1+1 = 4 - 4 = 0$$

$$\Rightarrow (10 \cdot 1)_4 = (10 \cdot 1)_2$$

$$= (0100 \cdot 01)_2 = (100 \cdot 01)_2$$

③ Binary Addition :- (★★★)

$$\begin{array}{r} (a) \quad 10 \ 11 \\ (+) \quad 01 \ 00 \\ \hline 11 \ 11 \end{array}$$

$$\begin{array}{r} (b) \quad 1 \ 1 \\ 1 \ 1 \ 1 \ 0 \ 11 \\ (+) \quad 1 \ 1 \ 1 \ 0 \ 00 \\ \hline 1 \ 1 \ 1 \ 0 \ 0 \ 11 \end{array}$$

$$1+1 = 2 - 2 = 0$$

$$1+1+1 = 3 - 2 = 1$$

$$\begin{array}{r} (c) \quad 0 \ 0 \ 11 \\ (+) \quad 0 \ 1 \ 0 \ 1 \\ \hline 1 \ 0 \ 0 \ 0 \end{array}$$

$$1+1 = 2 - 2 = 0$$

$$1+1+0 = 2 - 2 = 0$$

$$1+0+1 = 2 - 2 = 0$$

$$\begin{array}{r} (d) \quad 1 \ 1 \ 1 \ 1 \\ (+) \quad 1 \ 1 \ 1 \ 1 \\ \hline 1 \ 1 \ 1 \ 1 \ 0 \end{array}$$

$$1+1 = 2 - 2 = 0$$

$$1+1+1 = 3 - 2 = 1$$

$$1+1+1 = 3 - 2 = 1$$

$$1+1+1 = 3 - 2 = 1$$

④ Binary Subtraction :-

$$\begin{array}{r} (a) \quad 1001011011 \\ (-) \quad 10010000 \\ \hline 001000 \end{array}$$

$$\begin{array}{r} (b) \quad 101010 \\ (-) \quad 10110 \\ \hline 001000 \end{array}$$

$$\begin{array}{r} (c) \quad 1101010 \\ (-) \quad 0110000 \\ \hline 0111010 \end{array}$$

⑤ Binary Multiplication :-

$$\begin{array}{r} (a) \quad 1010 \\ \times 101 \\ \hline 1010 \\ 0000 \\ 1010 \\ \hline 110010 \end{array}$$

$$\begin{array}{r} (b) \quad 1111 \\ \times 1111 \\ \hline 1111 \\ 1111 \\ 1111 \\ 1111 \\ \hline 11110001 \end{array}$$

$$3+1+1+1 = 6$$

$$= (110)_2$$

$$3+1+1 = 5$$

$$= (101)_2$$

$$2+1 = 3$$

$$= 11$$

$$1+1+1+1 = 4$$

$$= (100)_2$$

$$1+1+1+1+1 = 5$$

$$= (101)_2$$

④ Binary division :-

(*) decimal division

$$\begin{array}{r} 4824 \div 6 = ? \\ 6 \overline{) 4824} \\ \underline{48} \\ 00 \\ \underline{00} \\ 0000 \end{array}$$

$$\begin{array}{r} 4824 \div 8 = ? \\ 8 \overline{) 4824} \\ \underline{48} \\ 24 \\ \underline{24} \\ 0000 \end{array}$$

$$\begin{array}{r} 11 \overline{) 110111001} \\ \underline{11} \\ 00011 \\ \underline{00011} \\ 00000 \\ \underline{00000} \\ 00000 \\ \underline{00000} \\ 00000 \end{array}$$

$$\begin{array}{r} 0101011 \\ 0000110 \\ \hline 0101100 \end{array}$$

(Q) $(723)_8 - (564)_8 = (?)_8$

$$\begin{array}{r} 723 \\ - 564 \\ \hline 137 \end{array}$$

(Q) $[ADD]_{16} + [DAD]_{16} = (?)_{16}$

$$\begin{array}{r} A D D \\ + D A D \\ \hline 188A \end{array} \approx (1)_{16}$$

$$\begin{aligned} 13+13 &= 26-16=10(A) \\ 13+10+1 &= 24-16=8 \\ 13+10+1 &= 24-16=8 \end{aligned}$$

$$\Rightarrow [000100010001010]_2$$

$$\Rightarrow \text{No. of 1's} = 5$$

(a) $(974B)_{16} - (587C)_{16} = (?)_{16}$

$$\begin{array}{r} 974B \\ - 587C \\ \hline 3ECF \end{array}$$

$16+11=27-12=15=F$
 $19-7=12=C$
 $22-8=14=E$

(Q) $(10W1Z)_2 \times (15)_{10} = (Y01011001)_2$; $W, Y, Z = ?$

A. M-I $[Z + 2 + 4W + 16] \times 15 = [256Y + 64 + 16 + 8 + 1]$

$$\Rightarrow 15Z + 30 + 60W + 240 = 256Y + 89$$

$$\Rightarrow 60W + 15Z + 270 = 256Y + 89 \quad [\text{trial \& errors}]$$

$$\Rightarrow W = Y = Z = 1$$

M-II

$$\begin{array}{r} 10W1Z \\ \times 1111 \\ \hline \end{array}$$

$$\begin{array}{r} 10W1Z \\ 10W1Z \\ 10W1Z \\ 10W1Z \\ \hline \end{array} \quad Z=1$$

$$\approx (Y01011001)_2$$

Codes

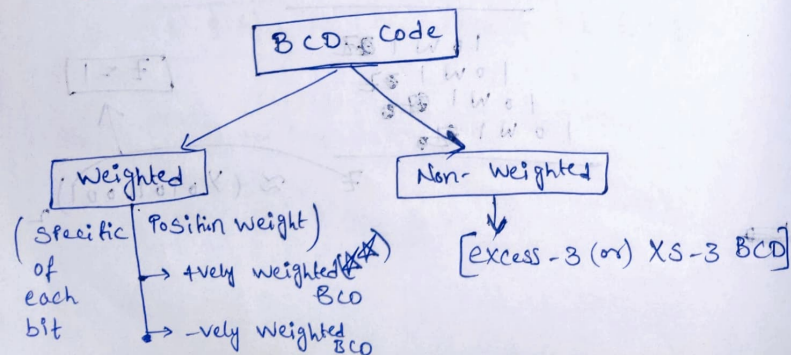
Eq: - FISH = # * 2.5 + 521

GATE = 2745

In digital design, we have binary, non-binary and alphanumeric codes.

Binary Codes

① BCD Codes:- [Binary Coded Decimal]



(*) If direct "BCD code" is written Consider it to be 8421 code.

$\Rightarrow$ Ex: $0(7)_{10} = (?)_{BCD} = (0111)_{BCD}$

$$\textcircled{2} (15)_{10} = (0001\ 0101)_{BCD}$$
$$(319)_{10} = (0011 \cdot 0001 \cdot 1001)_{8,CD}$$

② Excess (or) XS -3 Code :- It's a 4-bit code which can be derived from BCD code by adding "3" to each coded numbers.

Eg: $(0)_{10} = (1)_{x6-3}$

$$0 + 3 = 3$$

(0011) XS-3

② $(532)_{10} = (?)_{x5-3}$

$$2-3) \Rightarrow (865)_{10} = (1000 \ 0110 \ 0101)_{XS-3}$$

③ $(27)_{10} = (?)_{x_5-3}$

$$\Rightarrow (510)_6 = (0101 \overset{1010}{\cancel{1010}} \dots)_{XS-3}$$

① $(99)_{10} = \text{~~(1001 1001)~~ XS-3}$

↓ ↓

$12 \ 12 = (1100 \ 1100)_{XS-3}$

★ Binary v/s BCD

$(10)_{10} = (1010)_2 = (0001 \ 0000)_{BCD}$

★ All decimal digits in different BCD Codes :-

	8 4 2 1	XS-3	8 4 -2 -1
0	0 0 0 0	0 0 1 ←	0 0 0 0 ←
1	0 0 0 1	0 1 0 ←	0 1 1 1 ←
2	0 0 1 0	0 1 0 1 ←	0 1 1 0 ←
3	0 0 1 1	0 1 1 0 ←	0 1 0 1 ←
4	0 1 0 0	0 1 1 1 ←	0 1 0 0 ←
5	0 1 0 1	1 0 0 0 ←	1 0 1 1 ←
6	0 1 1 0	1 0 0 1 ←	1 0 1 0 ←
7	0 1 1 1	1 0 1 0 ←	1 0 0 1 ←
8	1 0 0 0	1 0 1 1 ←	1 0 0 0 ←
9	1 0 0 1	1 1 0 0 ←	1 1 1 1 ←

Eg:- ① $(697)_{10}$

$8 \ 4 \ 2 \ 1 \rightarrow 0110 \ 1001 \ 0111$

$XS-3 \rightarrow (9 \ 12 \ 7)_{XS-3}$

$\rightarrow (1001 \ 1100 \ 1010)_{XS-3}$

$8 \ 4 \ -2 \ -1 \rightarrow (1010 \ 1111 \ 1001)$

② The state of 12-bit register is 1000 100 1011. What is its content if it represents.

(a) 3-decimal digits in BCD

(b) " " " " " " XS-3

(a) $(8 \ 9 \ 7)_{10}$

(b) ~~(10 11 12)~~ $(5 \ 6 \ 7)_{10} \begin{bmatrix} 8-3 \\ 9-3 \\ 7-3 \end{bmatrix}$

$\Rightarrow$ XS-3 & 8 4 -2 -1 are self complementary codes.

$\Rightarrow$ 8 4 2 1 codes are not self-complementary.

Weighted $\rightarrow 8 \ 4 \ 2 \ 1$; 8 4 -2 -1 ; Non-weighted $\rightarrow XS-3$

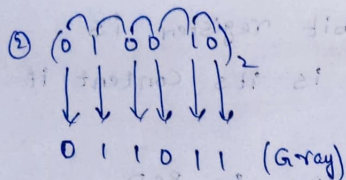
$\Rightarrow$ Gray Codes :- (unweighted codes) •
(unit distance codes) [K-Maps]

★ Binary to Gray conversion (B $\rightarrow$ G)

$(7)_{10} = (?)_{Gray}$

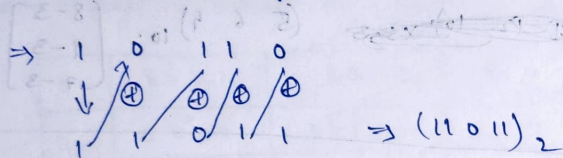
$(7)_{10} = (111)_2$

$\Rightarrow \begin{matrix} \oplus & \oplus \\ 1 & 1 & 1 \\ \downarrow & \downarrow & \downarrow \\ (1 \ 0 \ 0)_{Gray} \end{matrix}$



→ Gray to Binary :-

① (10 110) gray



② (10 111) gray ⇒ 1 0 1 1 1

1 1 0 1 0 ⇒ (11010)₂

D	Binary	Gray
0	0 0 0 0	0 0 0 0
1	0 0 0 1	0 0 0 1
2	0 0 1 0	0 0 1 1
3	0 0 1 1	0 0 1 0
4	0 1 0 0	0 1 1 0
5	0 1 0 1	0 1 1 1
6	0 1 1 0	0 1 0 1
7	0 1 1 1	0 1 0 0
8	1 0 0 0	1 1 0 0
9	1 0 0 1	1 1 0 1
10	1 0 1 0	1 1 1 1
11	1 0 1 1	1 1 1 0
12	1 1 0 0	1 0 1 0
13	1 1 0 1	1 0 1 1
14	1 1 1 0	1 0 0 1
15	1 1 1 1	1 0 0 0

(a) what all are unused cases of, 4-bits in BCD and XS-3 codes?

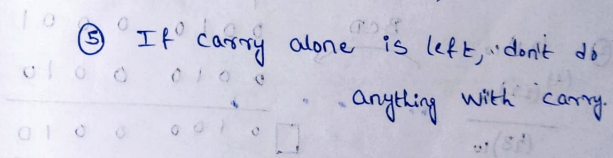
A.

unused

BCD	XS-3
10 10	0000
10 11	0001
1 1 0 0	0010
1 1 0 1	1101
1 1 1 0	1110
1 1 1 1	1111

⇒ BCD Addition :-

- Rules :-
- ① Add like binary.
 - ② $\text{Sum} \leq 9 \rightarrow \text{Valid}$
 - ③ $\text{Sum} \geq 9$ (or) Carry Present $\rightarrow \text{Invalid}$
 - ④ Valid $\rightarrow$ leave as it is
Invalid $\rightarrow$ Add 6 (0110)
 - ⑤ If carry alone is left, don't do anything with carry.



Procedure to Solve

Add like binary

(check the carry)

(Carry NOT generated)

Check Sum

> 9

Add 0110

< 9

(Leave)

(Carry

Generated)

Add (6)₁₀ = (0110)₂

[Don't do anything with

"left alone/Last" carry.]

Eg:-

$$\begin{array}{r} 0 \quad (2)_{10} \\ (+) \quad (3)_{10} \\ \hline (5)_{10} \end{array}$$

$$\begin{array}{r} \text{BCD} \\ 0010 \\ (+) 0011 \\ \hline 0101 \end{array}$$

$$\begin{array}{r} 2 \quad (21)_{10} \\ (+) \quad (22)_{10} \\ \hline (43)_{10} \end{array}$$

$$\begin{array}{r} \text{BCD} \\ 0010 \quad 0001 \\ (+) 0010 \quad 0010 \\ \hline 0100 \quad 0010 \end{array}$$

③

$$\begin{array}{r} (521)_{10} \\ (343)_{10} \\ \hline (864)_{10} \end{array}$$

$$\begin{array}{r} \text{BCD} \\ 0101 \quad 0010 \quad 0001 \\ (+) 0011 \quad 0100 \quad 0011 \\ \hline 1000 \quad 0110 \quad 0100 \end{array}$$

④

$$\begin{array}{r} 521.14 \\ 343.32 \\ \hline 864.46 \end{array}$$

$$\begin{array}{r} \text{BCD} \\ 0101 \quad 0010 \quad 0001 \quad 0100 \quad 0011 \\ (+) 0011 \quad 0100 \quad 0011 \quad 0011 \quad 0010 \\ \hline 1000 \quad 0110 \quad 0100 \quad 0111 \quad 0011 \end{array}$$

(*) Carry Involved :-

①

$$\begin{array}{r} 9 \\ (+) 8 \\ \hline 17 \end{array}$$

$$\begin{array}{r} 1001 \\ (+) 1000 \\ \hline 10001 \\ (+) 0110 \\ \hline 00010111 \\ [7]_{10} \end{array}$$

★ OBSERVE
Where to +6.
When the bit generate last carry, then should add. See ② & ③

②

$$\begin{array}{r} 92 \\ (+) 81 \\ \hline 173 \end{array}$$

$$\begin{array}{r} 1001 \quad 0010 \\ (+) 1000 \quad 0001 \\ \hline 10001 \quad 0011 \\ (+) 0110 \\ \hline 00010111 \quad 0011 \\ [1 \quad 7 \quad 3]_{10} \end{array}$$

$$\begin{aligned} & \text{ii) } (2, 5, 2) \\ & \text{iii) } (2, 5, 2) \\ & \text{iv) } (1, 1, 1) \end{aligned}$$

$\frac{11.152}{18.892}$
 0.5886

Carry forward

$$\begin{array}{r} P \\ 8 (+) \\ \hline F1 \end{array}$$